Sourceline Procurement System

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Sourceline is a procurement and stores system for a group of fifteen business units. It takes a purchase from the moment somebody asks for something to the moment the supplier is paid, and it refuses to let that happen out of order.

Each business unit runs on its own address with its own separate database. A person signing in at `khayah.sourceline.co.zw` is in Khayah Cement and can see nothing belonging to any other unit.

1. What problem it solves

Procurement spends money on behalf of someone else, which is why it is nearly always governed by a written procedure, and why the procedure is nearly always worked around when it lives in email and spreadsheets. The controls that matter are the dull ones: three quotes before an award, approval by the right people in the right order, and payment only against goods actually received.

Sourceline puts those controls in the path of the work rather than in a document beside it. The system will not let a purchase order reach a supplier without the approvals the value requires, and will not let an invoice be paid for goods nobody receipted.

2. What it does

Area	What it covers
Requisitions	Anyone can raise one; it is approved by their department head, checked against stock, then accepted by procurement
Sourcing	Requests for quotation to invited suppliers, quotations received and compared, award with a recorded justification
Purchase orders	Raised from the accepted quotation, approved through a chain that lengthens with value, then issued to the supplier
Goods receipt	Stores records what physically arrived, accepts it into stock or rejects it
Invoicing and payment	Supplier invoices against the order; a three-way match gates approval; payment is recorded
Suppliers	Self-registration, due diligence (KYS) by tier, document expiry tracking, bank-change control, periodic evaluation
Stores	Catalogue, stock on hand by site, issues, transfers between sites, reorder levels and low-stock alerts
Oversight	Budgets by department, an audit log of every action, dashboards per role, supplier performance reports

3. The controls

These are the parts worth understanding, because they are what the system is for.

Three quotations before an award. The default is three. Fewer is permitted only with a recorded waiver, and the waiver must name a reason from a fixed list — single source, approved contract, no quotes received, unique product, custom manufacture, and so on. The reason is kept with the award.

An approval chain that lengthens with value. A purchase order goes to the requesting department's head, then the head of Procurement, then Finance. Above USD 5,000 it also needs the COO. Each stage is a different person; skipping one is refused rather than warned about.

Three-way match before payment. An invoice is checked against the purchase order and against what stores actually receipted. Goods must have been received and accepted — not merely ordered — before finance can approve. VAT is reconciled separately. A variance can be overridden, but only with a written reason, which is kept on the invoice and in the audit log.

Supplier eligibility. A supplier may be invited to quote while its due diligence is still in progress, because asking for a price commits nothing. It cannot be *awarded* or issued a purchase order until its KYS is complete and it is marked spend-authorised. Diligence is tiered: a critical supplier carries the full checklist, a minor one a proportionate subset, and each tier has its own re-evaluation interval.

Separation between business units. Each unit has its own database. A session opened against one is refused by another. Document numbering restarts in each unit, so two units may both hold PO-2026-00001 — correct, not a collision.

4. The procurement cycle, step by step

#	Step	Who
1	Supplier registers, or is created by procurement	Supplier / Procurement officer
2	Due diligence completed and the supplier marked spend-authorised	Procurement officer
3	Requisition raised as a draft, then submitted	End user
4	Requisition approved	Department head
5	Checked against stock; issued from stores if available, otherwise forwarded	Stores officer
6	Requisition accepted for sourcing	Procurement officer
7	Request for quotation published to invited suppliers	Procurement officer
8	Quotations submitted	Suppliers
9	RFQ closed once quotes are in	Procurement officer
10	Winning quotation selected, with a written justification	Department head
11	Award authorised and accepted	Procurement officer
12	Purchase order raised from the accepted quotation and submitted	Procurement officer

#	Step	Who
13	Approved — requesting department head	Department head
14	Approved — head of Procurement	Procurement head
15	Approved — Finance	Finance
16	Authorised — COO, only at USD 5,000 or above	COO
17	Purchase order acknowledged	Supplier
18	Goods received; a goods received note is raised	Stores officer
19	Goods accepted into stock, or rejected	Stores officer
20	Invoice submitted against the order	Supplier
21	Three-way match; invoice approved	Finance
22	Payment recorded	Finance
23	Supplier evaluated and scored	Procurement officer

Steps 13 to 16 are the chain that lengthens with value. Step 16 is skipped below the threshold.

5. Who uses it

Role	What they do
End user	Raises requisitions for their department and tracks them
Department head	Approves their department's requisitions and purchase orders; selects the winning quotation
Procurement officer	Runs sourcing, awards, raises purchase orders, manages suppliers
Head of Procurement	A department head who also approves every purchase order at the second stage
Finance	Approves orders, approves invoices against the three-way match, records payments, manages budgets
COO	Authorises purchase orders at or above USD 5,000
Stores officer	Checks stock, receives goods, accepts or rejects, manages inventory and transfers
Supplier	Registers, completes due diligence, quotes, acknowledges orders, invoices

Role	What they do
Administrator	Manages users, departments and sites, and reads the audit log

6. How it is built

A TypeScript monorepo with three workspaces: the API, the web client, and a shared package of constants and rules used by both so the two cannot disagree about a status name or a threshold.

Layer	Technology
API	Node.js, Express 4, TypeScript 5.5
Database	MongoDB (Atlas), Mongoose 8
Web client	React 18, Vite 5, React Router 7, Tailwind CSS 3
Shared rules	A workspace package consumed by both sides
Authentication	JSON Web Tokens, bcrypt password hashing, a six-digit code as a second factor
Email	Resend
Documents	PDFKit for printed purchase orders and notes
Spreadsheets	SheetJS, for bulk supplier and stock import
Hosting	Vercel — the API as serverless functions, the client as a static build
Tooling	tsx, Prettier, concurrently, cross-env

Size. About 30,000 lines in the API across 291 files, 28,000 in the client across 107, and 1,400 in the shared package. 28 data models, 171 API endpoints, 42 screens.

7. How the multi-entity part works

The system began as one database for one company. It now serves fifteen, and the change had to be made without rewriting the 182 files that read and write data.

The registry. A central database holds one record per business unit: its code, its database name, its domains, its branding. Nothing else is shared.

Resolution by hostname. A request arriving at `dokuma.sourceline.co.zw` is matched against the registry, and everything that follows runs against Dokuma's database. The login page needs no chooser, because the address already said which unit.

Model resolvers. Each data model is a proxy that resolves, per request, to the copy bound to the current unit's database connection. Existing code was untouched; `PurchaseOrder.find()` means something different depending on who is asking.

Fail closed. Using a model with no business unit in context raises an error rather than reading whichever database happens to be open. There is no "all units" query — group-wide reporting has to fan out

deliberately.

Sessions are bound to their unit. A token issued for one unit is refused by another, so a session cannot be carried across by changing the address.

8. Running and operating it

Command	What it does
`npm run dev`	API on :3001 and client on :5173, both watching
`npm run typecheck`	Type-checks all three workspaces
`npm run check:isolation`	Proves two business units cannot see each other's data
`npm run check:numbering`	Proves concurrent documents get distinct numbers
`npm run e2e:local -- --sbu CODE <url>`	The full 50-step procurement cycle against a live API
`npm run seed:demo:prod -- --sbu CODE`	Test accounts, suppliers, catalogue and stock
`npm run seed:sbu-registry:prod -- --all`	Registers every business unit
`npm run bootstrap:sbus:prod -- --with-indexes`	Creates a unit's database, indexes and first administrator
`npm run sbu:domains:prod -- --base <domain>`	Assigns each unit its hostname
`npm run sbu:capacity`	Reports the cluster's collection budget
`npm run deploy:production`	Deploys API and client

Background jobs run on a schedule and raise alerts for low stock, approaching RFQ deadlines, expiring supplier documents and suppliers due for re-evaluation.

9. What is deliberately not built yet

Not built	Consequence today
Group-wide reporting across units	No single view of spend across the group
A shared supplier portal	A supplier works with each unit separately
Per-unit branding on printed documents	Every document prints "Fossil Contracting"
An SBU prefix on document numbers	Numbers repeat across units
Currency on orders, invoices and payments	Only quotations carry a currency
Per-unit approval thresholds	USD 5,000 applies everywhere
Per-unit links in emails	Every email links to one fixed address

Two of these follow from a deliberate decision that business units share no data at all. That gives the strongest separation available, and it is why a supplier must be onboarded in each unit that trades with it.

10. Where things stand

The platform is deployed and working. All fifteen units serve over HTTPS on their own hostnames, sign-in works, and the complete procurement cycle has been run end to end against the live system — fifty steps, no failures. Business units cannot see each other's data.

What remains before the group can use it in earnest is set-up rather than software: fourteen of the fifteen units are still empty, the administrator accounts cannot yet receive email, and two-factor sign-in is switched off while that is true. The database tier also needs confirming, because a shared tier carries no automated backup and the system is about to hold real payment records.